A Global Sidorenko Inequality for 2-Subdivided Alternating Cycles

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Let H_ℓ be obtained from the alternating cycle of length 2ℓ by replacing every directed edge with a directed path of two edges. We prove that, for every $\ell \geq 1$, its homomorphism density in every weighted tournament is at least $2^{-4\ell}$. This gives a global affirmative answer to the local tournament Sidorenko question posed by He, Mani, Nie, Tung and Wei for the entire family. The lengths 4, 8 and 12 were already covered by earlier cycle results.

The argument reduces all lengths to a matrix inequality: if u is a real unit vector and B is real skew-symmetric, then $\|(uu^{\mathsf{T}}/2 + B)^2\|_{\mathrm{op}} \geq 1/4$. An explicit vector proves this inequality through a nonnegative residual identity. A second spectral direction gives a dimension-free forcing estimate: the excess density is at least $16^{-\ell}\|(A - J/2)/n\|_{\mathrm{op}}^{4\ell}$. Equality holds precisely for the constant weighted tournament, and a three-vertex family shows that the exponent 4ℓ is optimal.

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1 The family and its global inequality

A tournament assigns a direction to each pair of distinct vertices. A weighted tournament permits a probability for each direction: it is a real matrix $A \in [0,1]^{n\times n}$ satisfying $A + A^{\mathsf{T}} = J$, where J is the all-ones matrix. In particular $A_{ii} = 1/2$. For an oriented graph F, write

$$t_F(A) = \frac{1}{n^{|V(F)|}} \sum_{\phi: V(F)\to[n]} \prod_{(v,w)\in E(F)} A_{\phi(v),\phi(w)}. \tag{1}$$

The sum includes all vertex maps, allowing vertices to coincide. The constant matrix $J/2$ has density $2^{-|E(F)|}$. A tournament Sidorenko inequality asks whether this random benchmark is a lower bound.

For $\ell \geq 1$, define H_ℓ on cyclically ordered vertices $v_0, \dots, v_{4\ell-1}$ by the edge pattern

$$v_{4j} \longrightarrow v_{4j+1} \longrightarrow v_{4j+2} \longleftarrow v_{4j+3} \longleftarrow v_{4j+4} \qquad (0 \leq j < \ell),$$

with indices modulo 4ℓ. Thus its clockwise edge word is $(++--)^\ell$. This is the directed 2-subdivision of the alternating cycle of length 2ℓ; the case $\ell = 1$ is the four-cycle consisting of two directed two-edge paths with common endpoints.

Problem 3.23 of He, Mani, Nie, Tung and Wei [HMN+25] asks whether every member of this family is *locally tournament Sidorenko*. In their Definition 3.1, this means that for each fixed F there is $\varepsilon(F) > 0$ such that $t_F(A) \geq 2^{-|E(F)|}$ whenever the skew-symmetric perturbation $A - J/2$ has spectral radius at most $\varepsilon(F)n$. For a real skew-symmetric matrix, spectral radius and operator norm agree. We prove the inequality without this restriction.

Throughout, $\|\cdot\|_{\mathrm{op}}$ is the Euclidean operator norm. We order the eigenvalues of a real symmetric matrix nonincreasingly.

Theorem 1.1. *For every integer $\ell \geq 1$, every integer $n \geq 1$ and every weighted tournament $A \in [0,1]^{n\times n}$, set*

$$B = \frac{A - J/2}{n}, \qquad \beta = \|B\|_{\mathrm{op}}.$$

Then

$$t_{H_\ell}(A) - 16^{-\ell} \;\geq\; 16^{-\ell}\beta^{4\ell}. \tag{2}$$

In particular $t_{H_\ell}(A) \geq 16^{-\ell}$, with equality if and only if $A = J/2$. For each fixed ℓ, the exponent 4ℓ in a positive-constant lower bound in terms of β cannot be decreased.

The theorem settles the complete local question affirmatively for every ℓ. It also explains quantitatively how a density close to the benchmark forces the host close to the constant matrix. If $E = t_{H_\ell}(A) - 16^{-\ell}$, then

$$\frac{1}{n^2} \max_{S,T\subseteq[n]} \left| \sum_{i\in S,\, j\in T} (A_{ij} - 1/2) \right| \leq \beta \leq 2E^{1/(4\ell)}. \tag{3}$$

Indeed, the first inequality follows by applying the operator norm to the indicator vectors of S and T, and the second is (2). Thus one cycle density controls every rectangle sum of edge-weight deviations, a standard quantitative expression of random-like behavior.

1.1 Prior work and the point of the proof

Grzesik, Il'kovič, Kielak and Král' [GIKK23] study quasirandom-forcing orientations of cycles. Their Theorem 12 already covers the members of the present family of lengths 4, 8 and 12. Theorem 1.1 treats every length 4ℓ uniformly and supplies the explicit forcing estimate (2). The four-cycle also occurs as D_2 in Fox, Himwich, Mani and Zhou [FHMZ24, Lemma 4.10]. Their further parallel-path constructions differ from the serial cycle family here.

The asymmetric Sidorenko transfer in [FHMZ22, Theorem 1.5] and [FHMZ24, Theorem 4.2] assumes a homomorphism to a single directed edge. Every H_ℓ has a vertex with both positive indegree and positive outdegree, so that hypothesis is absent. The distinction matters even though the underlying undirected graph is an even cycle.

Our main step is a lower bound for the largest singular value of a matrix square. Normalize $T = A/n$. The cycle word gives exactly

$$t_{H_\ell}(A) = \operatorname{tr}\big[(T^2(T^{\mathsf{T}})^2)^\ell\big].$$

The matrix inside the power is positive semidefinite. Consequently a single lower bound $\|T^2\|_{\mathrm{op}} \geq 1/4$ proves the density bound at every length. We obtain it from a vector adapted to the action of the skew part on the constant vector. The forcing estimate additionally controls a second eigenvalue.

Matrices with a rank-one symmetric part are classical almost skew-symmetric matrices. We use the block reduction of McDonald, Psarrakos and Tsatsomeros [MPT04, Equation (3.2)]. Their Theorem 3.7 includes a largest-singular-value lower bound for the matrix itself. A lower bound for $\|T\|_{\mathrm{op}}$ does not imply one for $\|T^2\|_{\mathrm{op}}$ by submultiplicativity, and T^2 need not have a rank-one symmetric part. For example, with $u = (1,0)^{\mathsf{T}}$ and $B = \left(\begin{smallmatrix} 0 & -1/4 \\ 1/4 & 0 \end{smallmatrix}\right)$, the symmetric part of $(uu^{\mathsf{T}}/2 + B)^2$ is $\operatorname{diag}(3/16, -1/16)$. The square estimate below is therefore proved directly.

Exact rational computations and Codex-assisted derivations were used in developing and checking the argument. The proofs below are analytic and self-contained; the computations are supplementary. The scope of this assistance is described at the end of the paper.

2 A lower bound for the matrix square

Lemma 2.1. *Let $u \in \mathbb{R}^n$ be a unit vector, $B^{\mathsf{T}} = -B$, $P = uu^{\mathsf{T}}$ and $T = P/2 + B$. Put $b = Bu$, $a = \|b\|^2$, and let D be the compression of B to $u^\perp$. For*

$$Q = T^2(T^{\mathsf{T}})^2$$

one has

$$\lambda_1(Q) \geq \frac{1}{16} + \frac{2a^2}{1+4a}. \tag{4}$$

In particular $\|T^2\|_{\mathrm{op}} \geq 1/4$, without any bound on B.

Proof. Skew-symmetry gives $b \perp u$ and $D^{\mathsf{T}} = -D$. In the orthogonal decomposition $\mathbb{R}u \oplus u^\perp$, the matrices are

$$B = \begin{pmatrix} 0 & -b^{\mathsf{T}} \\ b & D \end{pmatrix}, \qquad T = \begin{pmatrix} 1/2 & -b^{\mathsf{T}} \\ b & D \end{pmatrix}.$$

Set $x = u + 2b$. Then $\|x\|^2 = 1 + 4a > 0$. Two multiplications give

$$T^{\mathsf{T}}x = (1/2 + 2a)u - b - 2Db,$$
$$(T^{\mathsf{T}})^2 x = u/4 - (1/2 + 2a)b + Db + 2D^2 b.$$

Here the u-coordinate in the second line is exactly $1/4$, since $b^{\mathsf{T}}Db = 0$. The identities

$$\langle b, Db\rangle = \langle Db, D^2 b\rangle = 0, \qquad \langle b, D^2 b\rangle = -\|Db\|^2$$

yield the residual formula

$$\|(T^{\mathsf{T}})^2 x\|^2 - \frac{\|x\|^2}{16} = 2a^2 + 4a^3 + (3+8a)\|Db\|^2 + 4\|D^2 b\|^2. \tag{5}$$

For completeness, writing $c = 1/2 + 2a$, the squared norm on the left before subtraction is

$$\frac{1}{16} + c^2 a + (1 + 4c)\|Db\|^2 + 4\|D^2 b\|^2.$$

Subtracting $(1 + 4a)/16$ gives (5). Every residual term is nonnegative. Since $x^{\mathsf{T}} Q x = \|(T^{\mathsf{T}})^2 x\|^2$, the Rayleigh principle, followed by division by $1 + 4a$, proves (4). Finally $Q = T^2 (T^2)^{\mathsf{T}}$, so $\lambda_1(Q) = \|T^2\|_{\mathrm{op}}^2$. Empty blocks in dimension one are interpreted as zero. □

3 From the square to every cycle length

Take $u = n^{-1/2}(1, \dots, 1)^{\mathsf{T}}$ and $B = (A - J/2)/n$. Then $P = J/n$, $B^{\mathsf{T}} = -B$ and $T = A/n = P/2 + B$. Expanding a cyclic matrix product gives

$$t_{H_\ell}(A) = \operatorname{tr}\left[(TTT^{\mathsf{T}}T^{\mathsf{T}})^\ell\right] = \operatorname{tr}(Q^\ell). \tag{6}$$

In fact, at a clockwise edge the factor is T_{ij}, and at a counterclockwise edge it is $(T^{\mathsf{T}})_{ij} = T_{ji}$. Summing the cyclic indices is the trace. There are 4ℓ factors of A/n, giving exactly the normalization in (1), including coincident vertex maps.

All eigenvalues of Q are nonnegative. Lemma 2.1 therefore gives, for every n and every $\ell \geq 1$,

$$t_{H_\ell}(A) = \sum_{j=1}^{n} \lambda_j(Q)^\ell \geq \lambda_1(Q)^\ell \geq 16^{-\ell}. \tag{7}$$

This proves the affirmative answer to Problem 3.23 of [HMN+25] throughout the class of weighted tournaments.

4 Quantitative forcing and equality

We next extract a second spectral direction from the skew compression. Keep the notation of Lemma 2.1, and put $d = \|D\|_{\mathrm{op}}$.

Lemma 4.1. *If $n \geq 2$, then $\lambda_2(Q) \geq d^4$.*

Proof. Multiplying the block matrix for T gives

$$T^2 = \begin{pmatrix} 1/4 - a & -b^{\mathsf{T}}/2 - b^{\mathsf{T}} D \\ b/2 + Db & D^2 - bb^{\mathsf{T}} \end{pmatrix}.$$

Set $y = b/2 + Db$ and $K = -D^2 + bb^{\mathsf{T}}$. The compression of $Q = T^2(T^2)^{\mathsf{T}}$ to $u^\perp$ is

$$yy^{\mathsf{T}} + K^2, \qquad K \succeq -D^2 \succeq 0.$$

If $d = 0$, positivity of Q suffices. If $d > 0$, the largest eigenvalue d^2 of $-D^2$ has multiplicity at least two, by the real skew-symmetric block form. Let $F \subseteq u^\perp$ be its eigenspace. For every $z \in F$,

$$z^{\mathsf{T}} K z \geq d^2 \|z\|^2, \qquad \|Kz\| \geq d^2 \|z\|.$$

The second inequality follows from Cauchy–Schwarz (and is immediate for $z = 0$). Hence, viewing F as a subspace of $\mathbb{R}^n$,

$$z^{\mathsf{T}} Q z = (y^{\mathsf{T}} z)^2 + \|Kz\|^2 \geq d^4 \|z\|^2.$$

As $\dim F \geq 2$, the min-max principle gives $\lambda_2(Q) \geq d^4$. □

Proof of Theorem 1.1, except exponent optimality. Write $E = t_{H_\ell}(A) - 16^{-\ell}$. Dimension one has $A = J/2$, so suppose $n \geq 2$. By (6), Lemmas 2.1 and 4.1, and the elementary inequality $(s+h)^\ell \geq s^\ell + \ell s^{\ell-1}h$ for $s, h \geq 0$, we have both

$$E \geq \ell 16^{-(\ell-1)} \frac{2a^2}{1+4a}, \qquad E \geq d^{4\ell}. \tag{8}$$

For the second bound, retain $\lambda_1(Q)^\ell$ and $\lambda_2(Q)^\ell$ in the trace and use $\lambda_1(Q) \geq 1/16$.

Every entry of B has absolute value at most $1/(2n)$. Its Hilbert–Schmidt norm therefore gives

$$\beta \leq \|B\|_{\mathrm{HS}} \leq 1/2, \qquad a \leq \beta^2 \leq 1/4.$$

Decompose B into its extended compression D and the off-diagonal block $\left(\begin{smallmatrix} 0 & -b^{\mathsf{T}} \\ b & 0 \end{smallmatrix}\right)$. The latter has norm $\|b\| = \sqrt{a}$, so

$$\beta \leq d + \sqrt{a}. \tag{9}$$

If $d \geq \beta/2$, the second bound in (8) gives

$$E \geq (\beta/2)^{4\ell} = 16^{-\ell}\beta^{4\ell}.$$

Otherwise (9) gives $\sqrt{a} > \beta/2$. Since $2/(1+4a) \geq 1$, the first bound in (8) yields

$$E \geq \ell 16^{-(\ell-1)} a^2 \geq \ell 16^{-\ell}\beta^4 \geq 16^{-\ell}\beta^{4\ell}.$$

The last step uses $\ell \geq 1$ and $0 \leq \beta \leq 1$. This proves (2). If $E = 0$, then (2) forces $\beta = 0$, hence $A = J/2$. Conversely, that matrix has density $16^{-\ell}$ directly from (1). □

5 The forcing exponent is optimal

For $0 < r \leq 1/2$, consider the three-vertex weighted tournament

$$A = J/2 + rK_0, \qquad K_0 = \begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}.$$

Here $B = rK_0/3$ annihilates $u = 3^{-1/2}(1,1,1)^{\mathsf{T}}$, so $PB = BP = 0$. Its nonzero eigenvalues are $\pm ir/\sqrt{3}$, and $\beta = r/\sqrt{3}$. Consequently

$$T^2 = P/4 + B^2, \qquad Q = P/16 + B^4.$$

The eigenvalues of Q are $1/16$, β^4, β^4, and thus

$$t_{H_\ell}(A) - 16^{-\ell} = 2\beta^{4\ell}. \tag{10}$$

For each fixed ℓ and each $p < 4\ell$, the ratio of the left side to β^p tends to zero as $r \to 0$. No constant $c > 0$ can therefore make $E \geq c\beta^p$ valid for all weighted tournaments. This proves the exponent assertion of Theorem 1.1.

AI methodology

OpenAI Codex materially assisted problem selection, proof development, exact rational checks, literature comparison and writing throughout the paper. Separate Codex contexts also scrutinized the mathematical argument.

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